



**BAULKHAM HILLS HIGH SCHOOL  
YEAR 11 ASSESSMENT TASK 1 2024  
MATHEMATICS ADVANCED**

**MARKING COVER SHEET**

**FINAL MARK**

**Student Name:** \_\_\_\_\_ **Teacher:** \_\_\_\_\_

|                        | <b>Outcomes</b>     |                      |              |
|------------------------|---------------------|----------------------|--------------|
| <b>Page</b>            | <b>Functions F1</b> | <b>Statistics S1</b> | <b>Total</b> |
| <b>Multiple Choice</b> | Q1-4, 6, 7, 9<br>/7 | Q5, 8, 10<br>/3      | <b>/10</b>   |
| <b>Page 9</b>          | Q11<br>/5           |                      | <b>/5</b>    |
| <b>Page 10</b>         | Q12-14<br>/6        |                      | <b>/6</b>    |
| <b>Page 11</b>         | Q16<br>/1           | Q15<br>/2            | <b>/3</b>    |
| <b>Page 12</b>         | Q17-19<br>/5        |                      | <b>/5</b>    |
| <b>Page 13</b>         | Q20<br>/4           |                      | <b>/4</b>    |
| <b>Page 14</b>         | Q21-22<br>/5        |                      | <b>/5</b>    |
| <b>Page 15</b>         | Q24<br>/1           | Q23<br>/4            | <b>/5</b>    |
| <b>Page 16</b>         | Q25-26<br>/4        |                      | <b>/4</b>    |
| <b>Page 17</b>         | Q27<br>/3           |                      | <b>/3</b>    |
| <b>Page 18</b>         | Q29<br>/2           | Q28<br>/2            | <b>/4</b>    |
| <b>Page 19</b>         | Q30-31<br>/6        |                      | <b>/6</b>    |
| <b>Page 20</b>         | Q32-33<br>/5        |                      | <b>/5</b>    |
| <b>Page 21</b>         | Q35<br>/3           | Q34<br>/2            | <b>/5</b>    |
| <b>Pages 22-24</b>     | Q37-38<br>/6        | Q36<br>/3            | <b>/9</b>    |
| <b>TOTAL</b>           | <b>/63</b>          | <b>/16</b>           | <b>/79</b>   |

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**BAULKHAM HILLS HIGH SCHOOL**

**YEAR 11 ASSESSMENT TASK 1 2024**

# Mathematics Advanced

## General Instructions

- Reading time – 5 minutes
- Working time – 1 hour 30 minutes
- Write using black pen
- Board-approved calculators may be used
- Show all necessary working in Questions 11-38
- Marks may be deducted for careless or badly arranged work
- Reference sheet has been attached

**Total marks – 79**

**Exam consists of 24 pages.**

This paper consists of TWO sections.

## **Section I – (10 marks) Pages (4-7)**

### **Questions 1-10**

- Attempt Questions 1-10
- Answer on answer sheet provided
- Allow about 15 minutes for this section

## **Section II – (69 marks) Pages (9-24)**

- Attempt Questions 11-38
- Allow about 1 hour and 15 minutes

**Section I - 10 marks****Use the multiple choice answer sheet for questions 1-10**

1. Let  $f$  and  $g$  be two functions such that  $f(3) = 2$ ,  $f(5) = 3$ ,  $g(2) = 3$ ,  $g(4) = 5$ .

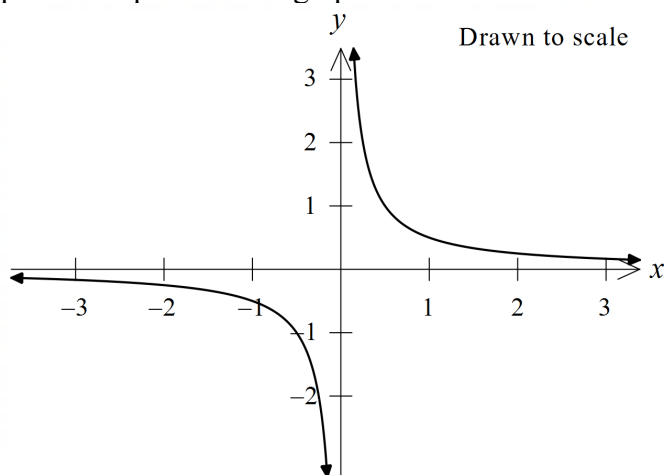
What is the value of  $f(g(4))$ ?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

2. Which of the following is equivalent to  $\frac{x^6}{y^6} \div x^2 y^{-3}$ ?

- (A)  $\frac{x^3}{y^2}$
- (B)  $\frac{x^4}{y^2}$
- (C)  $\frac{x^4}{y^3}$
- (D)  $\frac{x^4}{y^9}$

3. Which of the following equations represents the graph below?



- (A)  $y = \left(\frac{1}{2}\right)^x$
- (B)  $y = 2^x$
- (C)  $xy = \frac{1}{2}$
- (D)  $xy = 2$

4. What are the solutions of  $2x^2 - 5x - 1 = 0$ ?

(A)  $x = \frac{-5 \pm \sqrt{17}}{4}$

(B)  $x = \frac{5 \pm \sqrt{17}}{4}$

(C)  $x = \frac{-5 \pm \sqrt{33}}{4}$

(D)  $x = \frac{5 \pm \sqrt{33}}{4}$

5. Two hundred people were tested for COVID with the following results.

|                       | Tests positive | Tests negative |
|-----------------------|----------------|----------------|
| Showed symptoms       | 96             | 24             |
| Did not show symptoms | 8              | 72             |

What is the probability that the person tests positive to COVID, given that they didn't show symptoms?

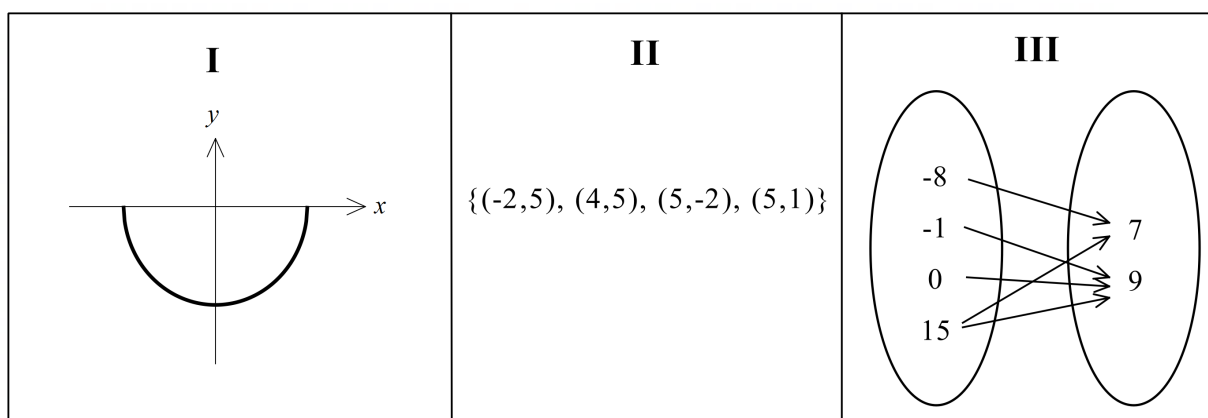
(A)  $\frac{1}{10}$

(B)  $\frac{1}{13}$

(C)  $\frac{1}{9}$

(D)  $\frac{1}{12}$

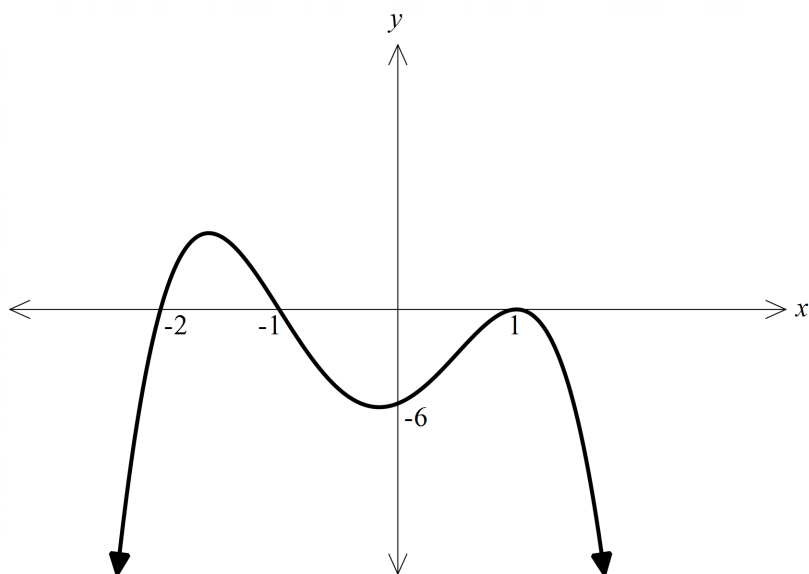
6. Which of the following represents a many-to-one relation?



- (A) I only  
 (B) I and II only  
 (C) I and III only  
 (D) I, II and III

7. The diagram shows the graph of  $y = a(x + b)(x + c)(x + d)^2$ .

What are possible values of  $a$ ,  $b$ ,  $c$  and  $d$ ?



- (A)  $a = -6$ ,  $b = -2$ ,  $c = -1$ ,  $d = 1$   
(B)  $a = -6$ ,  $b = 2$ ,  $c = 1$ ,  $d = -1$   
(C)  $a = -3$ ,  $b = -2$ ,  $c = -1$ ,  $d = 1$   
(D)  $a = -3$ ,  $b = 2$ ,  $c = 1$ ,  $d = -1$
8. Consider the two sets  $A = \{6, 7, 8\}$  and  $B = \{11, 12, 13\}$ . A number is randomly selected from each set.

What is the probability that the sum of the numbers selected is equal to or greater than 20?

- (A)  $\frac{1}{9}$   
(B)  $\frac{2}{9}$   
(C)  $\frac{1}{3}$   
(D)  $\frac{1}{6}$

9. Which of the following equations is **not** one-to-one?

(A)  $y = x^3$

(B)  $x^2 + y^2 = 4$

(C)  $y = 1 - 4x$

(D)  $y = \frac{4}{x}$

10. A game is played in which two different coloured dice are thrown once. The six faces of the red die are numbered 3, 5, 7, 8, 9 and 11. The six faces of the white die are numbered 1, 2, 4, 6, 10 and 12. The player wins if the number on the white die is larger than the number on the red die. What is the probability that the player wins exactly once in two successive games?

(A)  $\frac{7}{18}$

(B)  $\frac{11}{18}$

(C)  $\frac{77}{162}$

(D)  $\frac{77}{324}$

**Mathematics Advanced**  
**Section II Answer Booklet**

**69 marks**

**Attempt Questions 11-38**

**Allow about 1 hour and 15 minutes for this section**

**Instructions:**

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.

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**Please turn over**



**Question 11** (5 marks)**Marks**

Simplify the following:

a)  $\frac{18ab}{3a^2} \times \frac{c}{b}$

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b)  $\frac{x+2y}{3} - \frac{2x-y}{4}$

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c)  $\frac{(2x^{-4})^3}{5x^{-1}}$

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**Question 12** (2 marks)

A function is given by the formula  $f(x) = 4 \times 2^x$ .

a) Evaluate  $f(3)$ .

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b) Given that  $f(m) = 4$ , find the value of  $m$ .

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**Question 13** (2 marks)

Fully factorise  $3x^2 + 6x - 9$ .

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**Question 14** (2 marks)

Given the function  $f(x) = x^2 - 3x + 4$ , find and simplify  $f(\sqrt{a} + 2)$ .

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**Question 15** (2 marks)

Two bags contain a number of red and green marbles:

- bag A contains 3 red marbles and 4 green marbles.
- bag B contains 5 red marbles and 2 green marbles.

A six-sided die is rolled. If the uppermost face of the die shows a number that is 1 or 2, then a marble from bag A will be selected at random. Otherwise, a marble from bag B will be selected.

What is the probability that a green marble will be selected?

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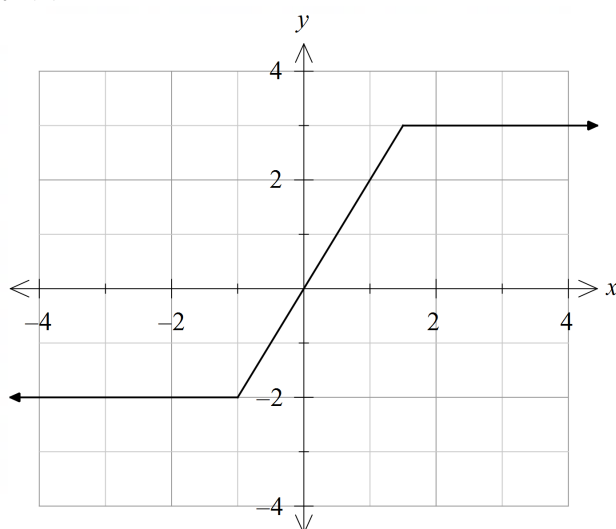
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**Question 16** (1 mark)

Consider the graph  $y = f(x)$  shown below.



Evaluate  $f(-3) + 2f(2)$ .

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**Question 17** (1 mark)

Write an equation of a straight line which is **not** a function.

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**Question 18** (2 marks)

Express  $(A - B)^2 + (A + B)^2$  in simplest form, if  $A^2 + B^2 = 3$ .

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**Question 19** (2 marks)

Solve  $\frac{7x}{3} = 1 + \frac{4x}{2}$ .

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**Question 20** (4 marks)

The perimeter of a rectangle is 34 cm and its area is  $72 \text{ cm}^2$ .

- a) If the width of the rectangle is  $x$  cm, show the area of the rectangle is  $A = 17x - x^2$ .

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- b) Hence find the length and width of the rectangle.

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**Question 21** (3 marks)

In 1 hour, 80 people attend a swimming pool and the money taken at the gate amounts to \$139. **3**

If the adults are charged \$2.50 and children \$1.50, how many of each entered the pool that hour?

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**Question 22** (2 marks)

The amount of petrol (P) used in a car is directly proportional to the distance (D) travelled. If **2**

the car uses 34.2 litres of petrol for an 85.5 km trip, find how far the car travelled if it used 35.5 L of petrol.

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**Question 23** (4 marks)

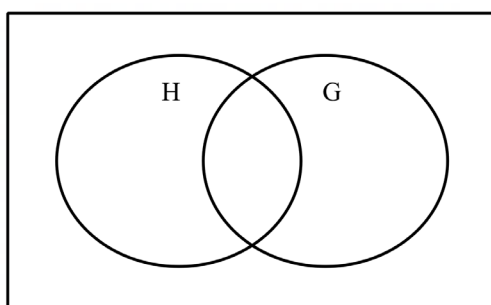
History (H) and Geography (G) are two of the subjects students may decide to study.

For a group of 40 students, the following is known.

- 7 students study neither History nor Geography
- 20 students study History
- 18 students study Geography

- a) A student is chosen at random. By completing the Venn diagram below, find the probability that the student studies both History and Geography.

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- b) Two different students are chosen at random, one after the other. What is the probability that the first student studies History and the second student does not study History?

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**Question 24** (1 mark)

Show that  $\frac{1}{3+\sqrt{3}} = \frac{3-\sqrt{3}}{6}$ .

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**Question 25** (2 marks)

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Solve the following simultaneous equations:

$$y = 2x + 3$$

$$xy = 2$$

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**Question 26** (2 marks)

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Simplify  $\frac{6^{3n} \times 9^n}{2^{3n}}$ .

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**Question 27** (3 marks)

Sketch the quadratic function  $y = -x^2 + 2x + 3$ , showing all important features.

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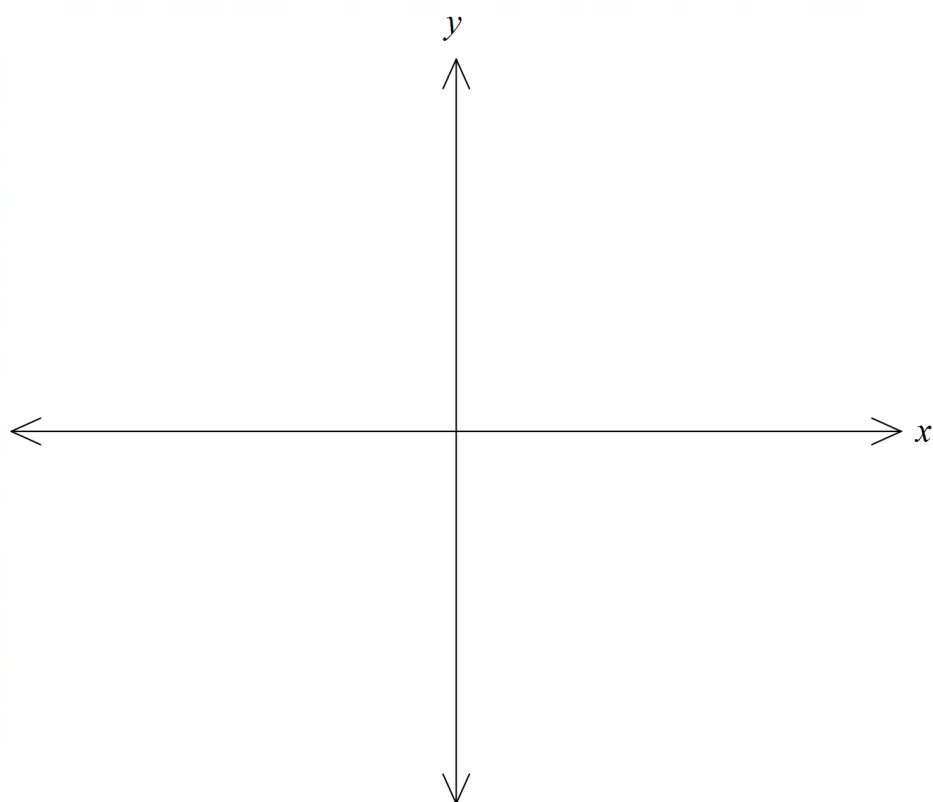
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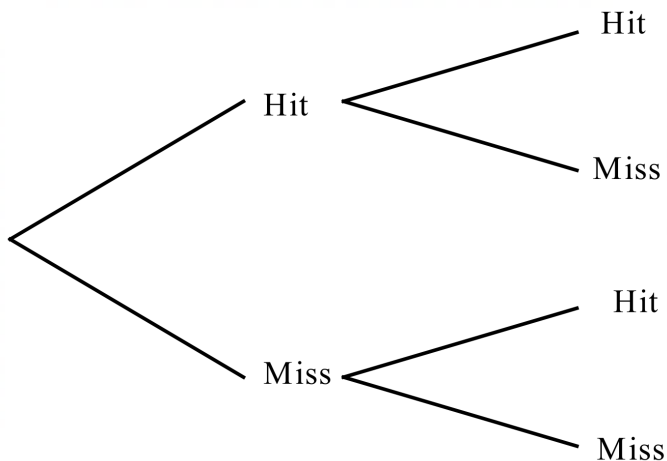
**Question 28** (2 marks)

Vaani is playing darts. She throws two darts aiming for a target.

The probability Vaani hits the target on her first throw is  $\frac{1}{4}$ . The probability she hits the target on her second throw is  $\frac{1}{3}$ .

- a) Complete the probability tree diagram showing the above information.

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- b) Find the probability that Vaani hits the target at least once.

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**Question 29** (2 marks)

Simplify  $\frac{x^2 - 9}{x^2 - x - 6} \div \frac{x^3 + 3x^2}{x^2 + 4x + 4}$ .

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**Question 30** (2 marks)

By completing the square, solve  $x^2 + 6x + 7 = 0$ . Express your answer in surd form.

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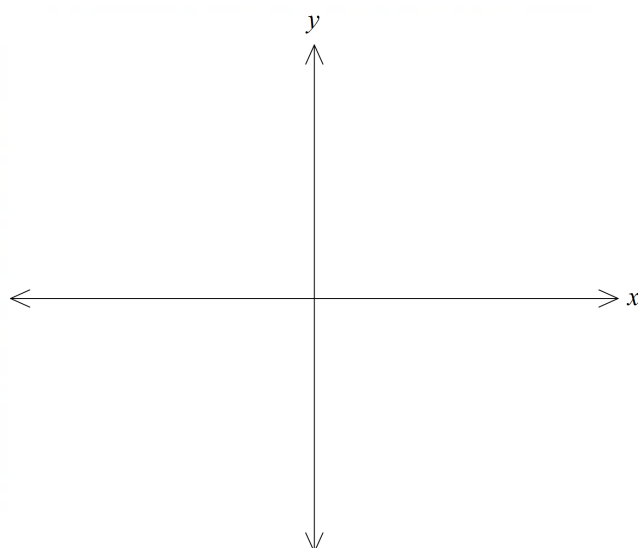
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**Question 31** (4 marks)

Consider  $f(x) = \sqrt{4 - x^2}$

a) Sketch  $f(x)$ .

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b) State the domain and range of  $f(x)$ .

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**Question 32** (2 marks)

For what value(s) of  $k$  does  $2x^2 - 3x - (k + 1) = 0$  have no real solutions?

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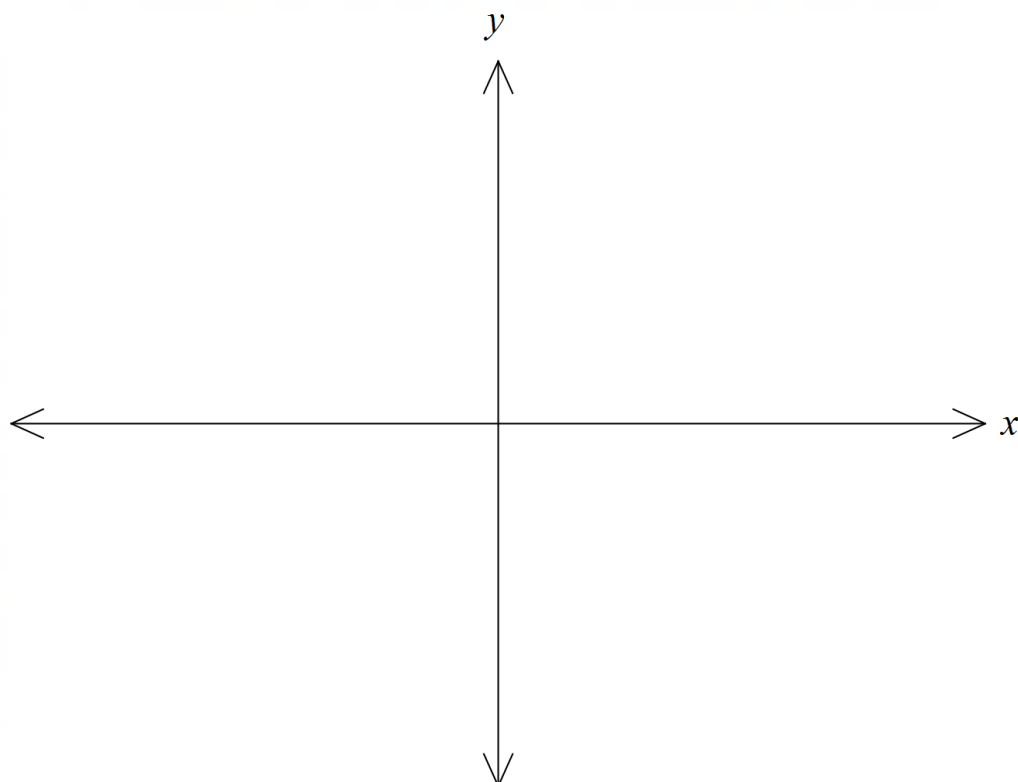
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**Question 33** (3 marks)

Sketch the graph of  $y = (x - 3)(x + 1)(x - 1)^3$ , showing all intercepts.

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**Question 34** (2 marks)

Two events,  $A$  and  $B$ , are such that  $P(A) = \frac{2}{7}$  and  $P(B) = \frac{3}{5}$ . Find the value of  $P(\overline{A} \cap B)$ , if

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$$P(A \cup B) = \frac{4}{5}.$$

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**Question 35** (3 marks)

Solve  $4y^4 - 5y^2 + 1 = 0$ .

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**Question 36** (3 marks)

A market research company is conducting a survey on health, diet, and exercise on the staff of a large factory.

They found that  $\frac{1}{3}$  of the factory workers took regular exercise and  $\frac{1}{2}$  of the factory workers ate breakfast. They further found that of those factory workers who took regular exercise  $\frac{1}{8}$  also ate breakfast.

A factory worker is selected at random.

- a) Find the probability that the worker does not eat breakfast and does not take regular exercise.

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- b) Determine, with reason, whether the events “take regular exercise” and “eat breakfast” are statistically independent.

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**Question 37 (3 marks)**

If  $\frac{2}{3x-2c} + \frac{3}{2x-3c} = \frac{7}{2c}$ , find  $x$  in terms of  $c$ .

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This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

**Question 38 (3 marks)**

The function  $f$  satisfies  $2f(x) + 3f\left(\frac{2x+3}{x-2}\right) = 3x+1$ ,  $x \in \mathbb{R}$ . Find the value of  $f(9)$ .

3

[illegible]

**End of Exam**





**BAULKHAM HILLS HIGH SCHOOL**  
**YEAR 11 ASSESSMENT TASK 1 2024**  
**MATHEMATICS ADVANCED**

Student Name: \_\_\_\_\_

Teacher: \_\_\_\_\_

**Mathematics**

**Section I – Multiple Choice**

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample:  $2 + 4 =$  (A) 2 (B) 6 (C) 8 (D) 9  
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐  
*correct*

- Start Here →
- |                            |                         |                         |                         |                             |                         |                         |                         |
|----------------------------|-------------------------|-------------------------|-------------------------|-----------------------------|-------------------------|-------------------------|-------------------------|
| 1. A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> | 6. A <input type="radio"/>  | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| 2. A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> | 7. A <input type="radio"/>  | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| 3. A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> | 8. A <input type="radio"/>  | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| 4. A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> | 9. A <input type="radio"/>  | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
| 5. A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> | 10. A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |

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Mathematics Advanced  
Mathematics Extension 1  
Mathematics Extension 2

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**REFERENCE SHEET****Measurement****Length**

$$l = \frac{\theta}{360} \times 2\pi r$$

**Area**

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

**Surface area**

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

**Volume**

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

**Functions**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For  $ax^3 + bx^2 + cx + d = 0$ :

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

**Relations**

$$(x - h)^2 + (y - k)^2 = r^2$$

**Financial Mathematics**

$$A = P(1 + r)^n$$

**Sequences and series**

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

**Logarithmic and Exponential Functions**

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

## Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

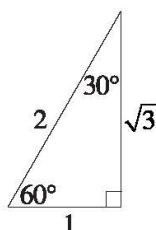
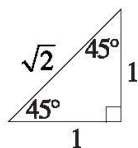
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



## Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

## Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

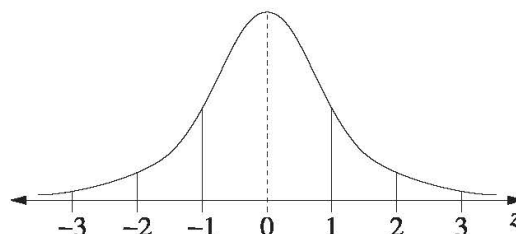
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

## Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score  
less than  $Q_1 - 1.5 \times IQR$   
or  
more than  $Q_3 + 1.5 \times IQR$

## Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

## Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

## Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

## Binomial distribution

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

# Year 11 Assessment Task 1 2024

## Mathematics Advanced

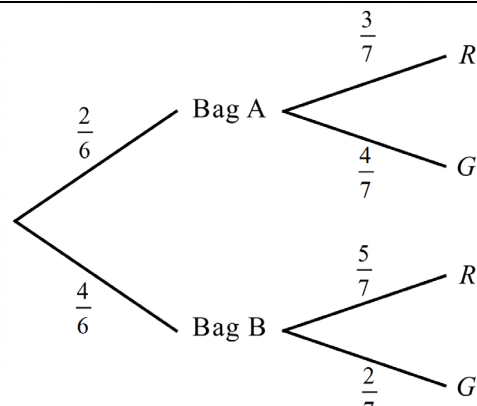
### Section I - Multiple Choice

1. B 2. C 3. C 4. D 5. A 6. A 7. D 8. C 9. B 10. C

| Question | Worked Solutions                                                                                                                                                                                               | Suggested Marking Criteria |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 1        | $f(g(4)) = f(5)$<br>$= 3$                                                                                                                                                                                      | B                          |
| 2        | $\frac{x^6}{y^6} \div x^2 y^{-3} = \frac{x^6}{y^6} \div \frac{x^2}{y^3}$ $= \frac{x^6}{y^6} \times \frac{y^3}{x^2}$ $= \frac{x^4}{y^3}$                                                                        | C                          |
| 3        | This is a hyperbola which eliminates options A and B.<br>Testing when $x = 1$ for option C, $y = 0.5$ .<br>Testing when $x = 1$ for option D, $y = 2$ .<br>Since the graph is drawn to scale, must be option C | C                          |
| 4        | $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $= \frac{-(-5) \pm \sqrt{(-5)^2 - 4(2)(-1)}}{2(2)}$ $= \frac{5 \pm \sqrt{33}}{4}$                                                                                     | D                          |
| 5        | $P(\text{tests positive}   \text{no symptoms}) = \frac{n(\text{test positive and no symptoms})}{n(\text{shows no symptoms})}$ $= \frac{8}{80}$ $= \frac{1}{10}$                                                | A                          |
| 6        | I is many-to-one.<br>II is many-to-many.<br>III is many-to-many.<br>$\therefore$ I only                                                                                                                        | A                          |

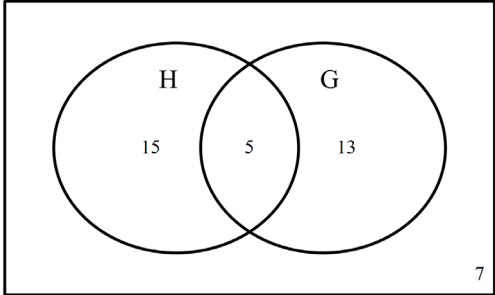
|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |    |           |    |   |    |    |    |   |    |    |    |   |    |    |    |    |         |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----|-----------|----|---|----|----|----|---|----|----|----|---|----|----|----|----|---------|---|--|--|--|--|--|--|--|---|--|--|--|--|--|--|--|---|--|--|--|--|--|--|--|---|--|--|--|--|--|--|--|---|--|--|--|--|--|--|--|----|--|--|--|--|--|--|--|---|
| 7       | $y = a(x + b)(x + c)(x + d)^2$ $y = a(x + 2)(x + 1)(x - 1)^2$ $-6 = a(0 + 2)(0 + 1)(0 - 1)^2$ $a = -3$ $\therefore a = -3, b = 2, c = 1, d = -1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | D         |    |           |    |   |    |    |    |   |    |    |    |   |    |    |    |    |         |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |
| 8       | <table border="1"><tr><td></td><td>11</td><td>12</td><td>13</td></tr><tr><td>6</td><td>17</td><td>18</td><td>19</td></tr><tr><td>7</td><td>18</td><td>19</td><td>20</td></tr><tr><td>8</td><td>19</td><td>20</td><td>21</td></tr></table> $P(\text{sum} \geq 20) = \frac{3}{9}$ $= \frac{1}{3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           | 11 | 12        | 13 | 6 | 17 | 18 | 19 | 7 | 18 | 19 | 20 | 8 | 19 | 20 | 21 | C  |         |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |
|         | 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 12        | 13 |           |    |   |    |    |    |   |    |    |    |   |    |    |    |    |         |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |
| 6       | 17                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 18        | 19 |           |    |   |    |    |    |   |    |    |    |   |    |    |    |    |         |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |
| 7       | 18                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 19        | 20 |           |    |   |    |    |    |   |    |    |    |   |    |    |    |    |         |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |
| 8       | 19                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 20        | 21 |           |    |   |    |    |    |   |    |    |    |   |    |    |    |    |         |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |
| 9       | Options A, C and D are all one-to-one.<br>$\therefore x^2 + y^2 = 4$ is many-to-many, <b>not</b> one-to-one                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | B         |    |           |    |   |    |    |    |   |    |    |    |   |    |    |    |    |         |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |
| 10      | <table border="1"><tr><td colspan="2"></td><td colspan="6">White die</td></tr><tr><td colspan="2"></td><td>+</td><td>1</td><td>2</td><td>4</td><td>6</td><td>10</td><td>12</td></tr><tr><td rowspan="6">Red die</td><td>3</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>5</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>7</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>8</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>9</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>11</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> $P(W) = \frac{14}{36}$ $\therefore P(LW) + P(WL) = \frac{22}{36} \times \frac{14}{36} + \frac{14}{36} \times \frac{22}{36}$ $= \frac{77}{162}$ |           |    | White die |    |   |    |    |    |   |    | +  | 1  | 2 | 4  | 6  | 10 | 12 | Red die | 3 |  |  |  |  |  |  |  | 5 |  |  |  |  |  |  |  | 7 |  |  |  |  |  |  |  | 8 |  |  |  |  |  |  |  | 9 |  |  |  |  |  |  |  | 11 |  |  |  |  |  |  |  | C |
|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | White die |    |           |    |   |    |    |    |   |    |    |    |   |    |    |    |    |         |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |
|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | +         | 1  | 2         | 4  | 6 | 10 | 12 |    |   |    |    |    |   |    |    |    |    |         |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |
| Red die | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |    |           |    |   |    |    |    |   |    |    |    |   |    |    |    |    |         |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |
|         | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |    |           |    |   |    |    |    |   |    |    |    |   |    |    |    |    |         |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |
|         | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |    |           |    |   |    |    |    |   |    |    |    |   |    |    |    |    |         |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |
|         | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |    |           |    |   |    |    |    |   |    |    |    |   |    |    |    |    |         |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |
|         | 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |    |           |    |   |    |    |    |   |    |    |    |   |    |    |    |    |         |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |
|         | 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |           |    |           |    |   |    |    |    |   |    |    |    |   |    |    |    |    |         |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |

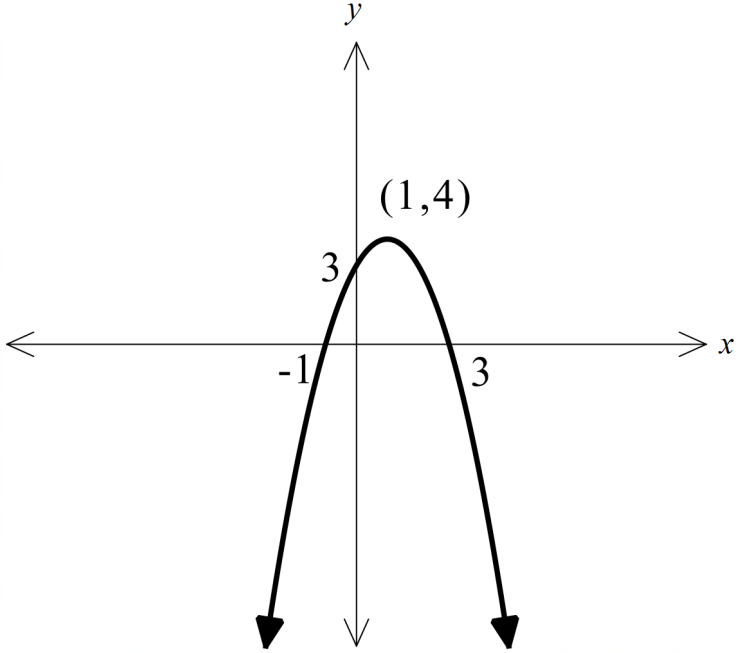
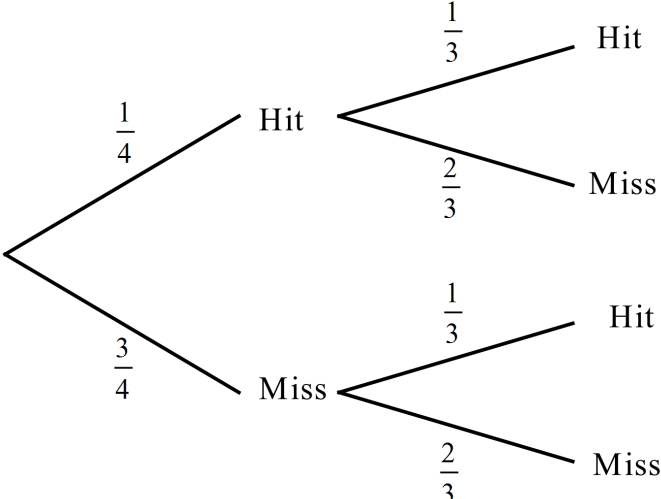
## Section II – Short answer questions

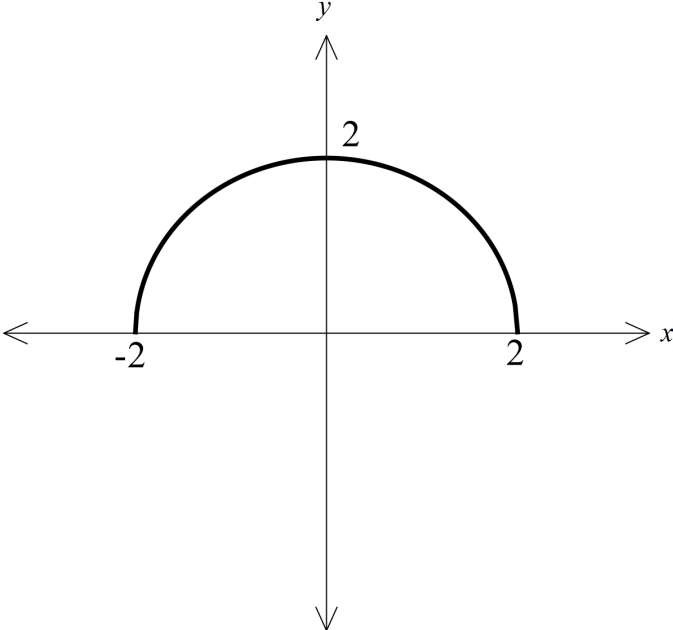
| Q   | Worked Solutions                                                                                                                                                                                                                                                                                                                                                                          | Suggested Marking Criteria                                                                                   |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 11a | $\frac{18ab}{3a^2} \times \frac{c}{b} = \frac{6c}{a}$                                                                                                                                                                                                                                                                                                                                     | 1 correct solution                                                                                           |
| 11b | $\frac{x+2y}{3} - \frac{2x-y}{4} = \frac{4(x+2y) - 3(2x-y)}{12}$ $= \frac{4x+8y-6x+3y}{12}$ $= \frac{-2x+11y}{12}$                                                                                                                                                                                                                                                                        | 2 correct solution<br>1 correct attempt to making the denominator the same                                   |
| 11c | $\frac{(2x^{-4})^3}{5x^{-1}} = \frac{8x^{-12}}{5x^{-1}}$ $= \frac{8x^{-11}}{5} \text{ or } \frac{8}{5x^{11}}$                                                                                                                                                                                                                                                                             | 2 correct solution<br>1 expanding the bracket                                                                |
| 12a | $f(3) = 4 \times 2^3$ $= 32$                                                                                                                                                                                                                                                                                                                                                              | 1 correct solution                                                                                           |
| 12b | $4 = 4 \times 2^m$ $1 = 2^m$ $\therefore m = 0$                                                                                                                                                                                                                                                                                                                                           | 1 correct solution                                                                                           |
| 13  | $3x^2 + 6x - 9 = 3(x^2 + 2x - 3)$ $= 3(x-1)(x+3)$                                                                                                                                                                                                                                                                                                                                         | 2 correct solution<br>1 correctly factorises 3 as a HCF                                                      |
| 14  | $f(\sqrt{a}+2) = (\sqrt{a}+2)^2 - 3(\sqrt{a}+2) + 4$ $= a + 4\sqrt{a} + 4 - 3\sqrt{a} - 6 + 4$ $= \sqrt{a} + a + 2$                                                                                                                                                                                                                                                                       | 2 correct solution<br>1 correctly expands the perfect square                                                 |
| 15  |  <p> <math display="block">\text{Bag A then Green} = \frac{2}{6} \times \frac{4}{7} = \frac{4}{21}</math> <math display="block">\text{Bag B then Green} = \frac{4}{6} \times \frac{2}{7} = \frac{4}{21}</math> <math display="block">\therefore \frac{4}{21} + \frac{4}{21} = \frac{8}{21}</math> </p> | 2 correct solution<br>1 correctly identifies the probability of obtaining bag A and the probability of bag B |

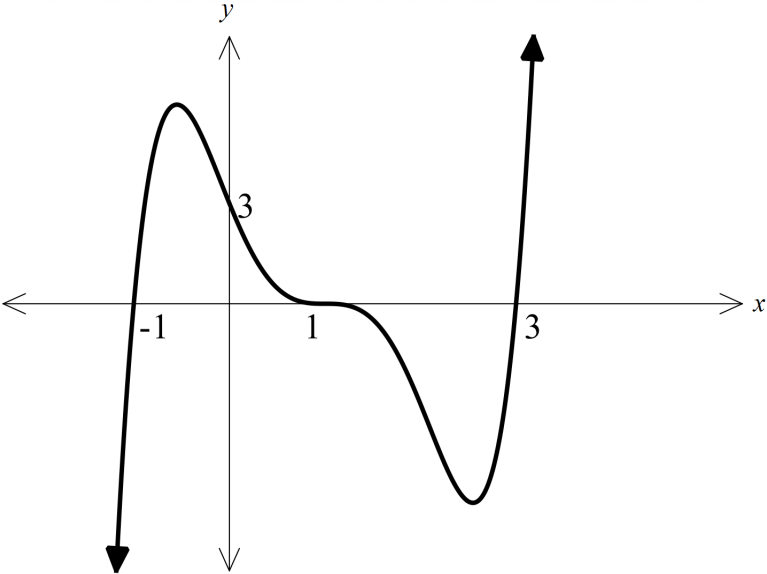
|            |                                                                                                                                                                                                                                    |                                                                                                                                      |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>16</b>  | $f(-3) = -2$<br>$f(2) = 3$<br>$\therefore f(-3) + 2f(2) = -2 + 6$<br>$= 4$                                                                                                                                                         | 1 correct solution                                                                                                                   |
| <b>17</b>  | $x = 2$ or $x = -2$ etc<br>Any vertical line                                                                                                                                                                                       | 1 correct solution                                                                                                                   |
| <b>18</b>  | $(A - B)^2 + (A + B)^2 = A^2 - 2AB + B^2 + A^2 + 2AB + B^2$<br>$= 2A^2 + 2B^2$<br>$= 2 \times 3$<br>$= 6$                                                                                                                          | 2 correct solution<br>1 correctly expands the expression                                                                             |
| <b>19</b>  | $\frac{7x}{3} = 1 + \frac{4x}{2}$<br>$14x = 6 + 12x$<br>$2x = 6$<br>$x = 3$                                                                                                                                                        | 2 correct solution<br>1 multiplies both sides by 6                                                                                   |
| <b>20a</b> | let length be $l$<br>$P = 2x + 2l$<br>$34 = 2x + 2l$<br>$l = 17 - x$<br><br>$A = xl$<br>$= x(17 - x)$<br>$= 17x - x^2$                                                                                                             | 2 correctly shows the area is $17x - x^2$<br>1 correctly shows the length is $17 - x$                                                |
| <b>20b</b> | $72 = 17x - x^2$<br>$x^2 - 17x + 72 = 0$<br>$(x - 9)(x - 8) = 0$<br>$x = 9$ or $8$<br>if $x = 9$ , $l = 8$<br>or<br>if $x = 8$ , $l = 9$                                                                                           | 2 correct solution<br>1 forms a quadratic to solve<br>Note: Does not need both solutions                                             |
| <b>21</b>  | let $a$ = adult and $c$ = children<br>$a + c = 80 \Rightarrow c = 80 - a$<br>$2.50a + 1.50c = 139$<br>$2.50a + 1.50(80 - a) = 139$<br>$2.50a + 120 - 1.50a = 139$<br>$a = 19$ , $c = 61$<br>$\therefore$ 19 adults and 61 children | 3 correct solution<br>2 correctly attempts simultaneous equations<br>1 correctly forms the 2 equations before simultaneous equations |



|     |                                                                                                                                                                                            |                                                                                                     |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 22  | $P = kD$<br>$34.2 = k \times 85.5$<br>$k = 0.4$<br>$35.5 = 0.4 \times D$<br>$D = 88.75 \text{ km}$                                                                                         | 2 correct solution<br>1 correctly finds the constant term                                           |
| 23a | <p>To find the intersection:<br/> <math>40 - 7 - 20 - 18 = 5</math></p>  <p><math>\frac{5}{40}</math></p> | 2 correct venn diagram and probability<br>1 correct venn diagram or probability                     |
| 23b | $\frac{20}{40} \times \frac{20}{39} = \frac{10}{39}$                                                                                                                                       | 2 correct solution<br>1 correctly identifies the probability that the first student studies History |
| 24  | $\frac{1}{3 + \sqrt{3}} \times \frac{3 - \sqrt{3}}{3 - \sqrt{3}} = \frac{3 - \sqrt{3}}{9 - 3}$ $= \frac{3 - \sqrt{3}}{6}$                                                                  | 1 shows multiplying the conjugate surd                                                              |
| 25  | $x(2x + 3) = 2$<br>$2x^2 + 3x - 2 = 0$<br>$(x + 2)(2x - 1) = 0$<br>$x = -2, \quad x = \frac{1}{2}$<br>if $x = -2, y = -1$<br>if $x = \frac{1}{2}, y = 4$                                   | 2 correct solutions<br>1 correctly substitutes and solves for $x$                                   |
| 26  | $\frac{6^{3n} \times 9^n}{2^{3n}} = \frac{(2 \times 3)^{3n} \times 3^{2n}}{2^{3n}}$ $= \frac{2^{3n} \times 3^{3n} \times 3^{2n}}{2^{3n}}$ $= 3^{5n}$                                       | 2 correct solution<br>1 correctly obtains $2^{3n} \times 3^{3n}$ for $6^{3n}$                       |

|     |                                                                                              |                                                                                                                                                                                                                                                                                                                                                       |
|-----|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 27  |            | <p>3 correct graph with all important features provided</p> <ul style="list-style-type: none"> <li>- <math>x</math> and <math>y</math> intercepts</li> <li>- Turning point</li> <li>- Correct shape</li> </ul> <p>2 correct <math>x</math> intercept and correct turning point</p> <p>1 correct <math>x</math> intercept or correct turning point</p> |
| 28a |            | 1 correct shows the probability on all branches                                                                                                                                                                                                                                                                                                       |
| 28b | $1 - (\text{miss both}) = 1 - \left( \frac{3}{4} \times \frac{2}{3} \right)$ $= \frac{1}{2}$ | 1 correct solution                                                                                                                                                                                                                                                                                                                                    |

|     |                                                                                                                                                         |                                                                                                                                                    |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 29  | $\frac{x^2 - 9}{x^2 - x - 6} \div \frac{x^3 + 3x^2}{x^2 + 4x + 4} = \frac{(x-3)(x+3)}{(x-3)(x+2)} \times \frac{(x+2)^2}{x^2(x+3)}$ $= \frac{x+2}{x^2}$  | 3 correct solution<br>2 correctly factorises three expressions and multiplies by the reciprocal<br>1 correctly factorises at least two expressions |
| 30  | $x^2 + 6x + 7 = 0$ $x^2 + 6x + \left(\frac{6}{2}\right)^2 - \left(\frac{6}{2}\right)^2 + 7 = 0$ $(x+3)^2 = 2$ $x+3 = \pm\sqrt{2}$ $x = -3 \pm \sqrt{2}$ | 2 correct solution<br>1 correctly forms a perfect square                                                                                           |
| 31a |                                                                      | 2 correct graph showing all important features<br>1 identifies it is a semi circle                                                                 |
| 31b | Domain: $-2 \leq x \leq 2$<br>Range: $0 \leq y \leq 2$                                                                                                  | 2 correct solution<br>1 correct domain or range                                                                                                    |

|    |                                                                                                                                                                                                   |                                                                                                                                 |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 32 | $b^2 - 4ac < 0$ $(-3)^2 - 4(2)(-k - 1) < 0$ $9 - 8(-k - 1) < 0$ $k < -\frac{17}{8}$                                                                                                               | 2 correct solution<br>1 identifies $\Delta < 0$                                                                                 |
| 33 |                                                                                                                | 3 correct graph<br>2 identifies a cubic root and has correct and has correct x and y intercepts<br>1 correct x and y intercepts |
| 34 | $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $\frac{4}{5} = \frac{2}{7} + \frac{3}{5} - P(A \cap B)$ $P(A \cap B) = \frac{3}{35}$ $P(\bar{A} \cap B) = \frac{3}{5} - \frac{3}{35}$ $= \frac{18}{35}$ | 2 correction solution<br>1 correctly finds P(A and B)                                                                           |
| 35 | let $x = y^2$<br>$4x^2 - 5x + 1 = 0$<br>$(4x - 1)(x - 1) = 0$<br>$x = \frac{1}{4}$ or $x = 1$<br>When $x = \frac{1}{4}, y = \pm \frac{1}{2}$<br>When $x = 1, y = \pm 1$                           | 3 correct solutions<br>2 correctly finds x or equivalent<br>1 correctly reduces the quartic to a quadratic equation             |

|     |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 36a | $P(E) = \frac{1}{3}$ $P(B) = \frac{1}{2}$ $P(B E) = \frac{1}{8}$ $P(B E) = \frac{P(B \cap E)}{P(E)}$ $\frac{1}{8} = \frac{P(B \cap E)}{\frac{1}{3}}$ $P(B \cap E) = \frac{1}{24}$ $\therefore P(\bar{B} \cap \bar{E}) = 1 - \left( \frac{1}{3} + \frac{1}{2} - \frac{1}{24} \right) = \frac{5}{24}$                                                                            | 2 correct solution<br>1 correctly finds $P(B \cap E)$<br><br>0 bold answer                                                                                     |
| 36b | <u>Method 1:</u><br>If independent: $P(E) \times P(B) = P(E \cap B)$<br>$P(E) \times P(B) = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$ $P(E \cap B) = \frac{1}{24}$ Since $P(E) \times P(B) \neq P(E \cap B)$<br>$\therefore$ Not independent events                                                                                                                        | 1 correct proof<br><br>(must identify rules used and clearly state values of $P(E)$ , $P(B)$ and $P(E \cap B)$ )                                               |
|     | <u>Method 2:</u><br>If independent: $P(B E) = P(B)$<br>$P(B E) = \frac{1}{8}$ $P(B) = \frac{1}{2}$ Since $P(B E) \neq P(B)$<br>$\therefore$ Not independent events                                                                                                                                                                                                             | 1 correct proof<br>(must identify rules used and clearly state values of $P(B)$ , $P(B E)$ )                                                                   |
| 37  | $\frac{2}{3x-2c} + \frac{3}{2x-3c} = \frac{7}{2c}$ $\frac{2(2x-3c) + 3(3x-2c)}{(3x-2c)(2x-3c)} = \frac{7}{2c}$ $\frac{4x-6c+9x-6c}{6x^2-9cx-4cx+6c^2} = \frac{7}{2c}$ $\frac{13x-12c}{6x^2-13cx+6c^2} = \frac{7}{2c}$ $2c(13x-12c) = 7(6x^2-13cx+6c^2)$ $26cx-24c^2 = 42x^2-91cx+42c^2$ $42x^2-117cx+66c^2 = 0$ $3(x-2c)(14x-11c) = 0$ $x = 2c \text{ or } x = \frac{11c}{14}$ | 3 correct solution<br>2 Finds a correct equation by substituting and and attempts to solve simultaneously<br><br>1 Finds a correct equation by substituting or |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                             |
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| 38 | <p>When <math>x = 9</math>,</p> $2f(9) + 3f\left(\frac{2(9)+3}{9-2}\right) = 3(9) + 1$ $2f(9) + 3f(3) = 28 \quad (1)$ <p>When <math>x = 3</math>,</p> $2f(3) + 3f\left(\frac{2(3)+3}{3-2}\right) = 3(3) + 1$ $2f(3) + 3f(9) = 10$ $f(3) = 5 - \frac{3}{2}f(9) \quad (2)$ <p>Sub (2) into (1)</p> $2f(9) + 3\left(5 - \frac{3}{2}f(9)\right) = 28$ $2f(9) + 15 - \frac{9}{2}f(9) = 28$ $-\frac{5}{2}f(9) = 13$ $\therefore f(9) = -\frac{26}{5}$ | <p>3 correct solution<br/> 2 finds an expression for <math>f(3)</math> and <math>f(9)</math>, and attempts to solve simultaneously<br/> 1 correctly finds <math>f(9)</math> in terms of <math>f(3)</math> or equivalent</p> |
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